Network Management Framework & SNMP Protocol

Coverage: Managing Entities, Agents, MIB Database, SNMP Protocol Operations, SNMPv1 vs SNMPv2c vs SNMPv3 (USM & VACM)

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1. SNMP Framework & Operations

- 4 Pillars: Manager (NMS), Agents, MIB, SNMP protocol (UDP 161/162).
- Operations: GetRequest, GetNextRequest, GetBulkRequest, SetRequest, Response, Trap, InformRequest.

2. SNMPv3 Security

- USM: HMAC-SHA authentication, AES encryption, replay protection.

■ KTU University High-Yield Exam Questions

- Q: Explain the SNMP architecture and operations. Compare SNMPv1, v2c, and v3. (8 Marks)